

Computer Networks

Lecture 24 Network Layer

Content



- **Classless Addressing**

Questions

1. An ISP is granted a block of addresses as 190.100.0.0/16. Design the blocks in which
 - a. 1st group has 64 customers and each customer requires 256 IP address
 - b. 2nd group has 128 customers and each customer requires 128 IP address
 - c. 3rd group has 128 customers and each customer requires 64 IP address
2. In the last problem how many address are still left over?
3. Given the network address of 112.44.0.0 where station s_1 has 112.44.22.19/16 and station s_2 112.44.23.2/16
 - a. Identify whether these 2 hosts belong to the same n/w or not
 - b. Calculate the last address of that network if both host belongs to same network.
4. Which of the following device share the same n/w
 - a. 192.168.78.25/29
 - b. 192.168.78.23/29
 - c. 192.168.78.33/29
 - d. 192.168.78.41/29

Questions

5. An ISP has the following chunk of CIDR-based Ip addresses available with it: 128.248.64.0/21. The ISP wants to give half of these addresses to organization A and the remaining half to organization B. What is the correct allocation of blocks to A and B, respectively?
6. Perform CIDR aggregation (also called supernetting)on the following IP addresses:
 - 128.56.24.0/24
 - 128.56.25.0/24
 - 128.56.26.0/24
 - 128.56.27.0/24
7. An IP router implementing CIDR receives a packet with the address of 131.23.151.76.The routing table is as follows:

Prefix	O/P Identifier
131.16.0.0/12	3
131.28.0.0/14	5
131.19.0.0/16	2
131.23.0.0 /15	1

The identifier of the output interface on which this packet will be forwarded is